

MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust ®, Mangalore)

Autonomous Institute Affiliated to V.T.U, Belagavi, Approved by AICTE, New Delhi
Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified Institution



PROJECT PHASE – 1 REPORT

“EDGE-INTELLIGENT CONDITION MONITORING SYSTEM”

**Submitted in partial fulfillment of the requirements for the award
of the degree of**

BACHELOR OF ENGINEERING

in

MECHATRONICS

by

**G Lakshman Shrestha
Anoop Kumar A S
Prajwal M
Rakesh Salian**

**4MT23MT012
4MT24MT402
4MT24EC404
4MT24EC407**

Under the Guidance of

**Dr. KiranKumar M V
Senior Assistant Professor
Dept. of Mechatronics**

**Dr. Srikrishna Shastri C
Associate Professor
Dept. of ECE**

DEPARTMENT OF MECHATRONICS

(Accredited by NBA)

2025 - 26

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CERTIFICATE

This is to certify that the project work entitled “**EDGE-INTELLIGENT CONDITION MONITORING SYSTEM**” is a bonafide work carried out by **Mr. G Lakshman Shrestra (4MT23MT012), Mr. Anoop Kumar A S (4MT24MT402), Mr. Prajwal M (4MT24EC404) & Mr. Rakesh Salian (4MT24EC407)** in partial fulfillment for the award of degree of **Bachelor of Engineering in Mechatronics** of **Mangalore Institute of Technology & Engineering (Autonomous)** during the year **2025 - 26**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

Dr. KiranKumar M V
Project Guide

Dr. Srikrishna Shastri C
Project Guide

Dr. Bindu Madhavi J
Head of Dept. Mechatronics

Dr. Vinayambika S Bhat
Head of Dept. ECE

Dr. Prashanth C M
Principal

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G Lakshman Shrestra
4MT23MT012

Anoop Kumar A S
4MT24MT402

Prajwal M
4MT24EC404

Rakesh Salian
4MT24EC407

ABSTRACT

This research presents a robust, noise-resilient Artificial Intelligence framework for industrial predictive maintenance utilizing Tiny Machine Learning deployed on low-power, dual-core microcontrollers. The primary objective involves shifting computational intelligence directly to the machine source to mitigate conventional cloud-based limitations, including high latency, excessive bandwidth consumption, and data privacy concerns. The methodology employs a multi-modal data fusion strategy, integrating triaxial vibration signals, non-invasive electrical current data, and thermal readings to provide a comprehensive health profile of rotating machinery. A noise-resilient preprocessing pipeline utilizing the Fast Fourier Transform converts raw time-domain signals into identifiable spectral signatures for classification by memory-optimized, 8-bit quantized convolutional neural networks. The experimental setup utilizes a variable frequency drive to control a three-phase induction motor, enabling diagnostic validation across various operational regimes. Results demonstrate diagnostic accuracies exceeding 92% with sub-second inference latency below 250 milliseconds. This decentralized approach is projected to reduce maintenance costs by 30% and unplanned downtime by 40% while achieving significant energy savings compared to cloud-streaming systems.

Keywords: Condition Monitoring, Edge Intelligence, Fast Fourier Transform, Predictive Maintenance, Sensor Fusion.

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CHAPTER 1

INTRODUCTION

This chapter provides the foundational context for the research, establishing the necessity of transitioning from traditional maintenance models to intelligent, edge-native predictive strategies. It defines the project's technical scope by outlining maintenance philosophies, the specific mechanisms of Edge-Intelligent Condition Monitoring system, work objectives, and the hardware and software prerequisites required for implementation.

1.1 Overview

The "Edge-Intelligent Condition Monitoring System" is a robust, noise-resilient Artificial Intelligence (AI) framework designed for industrial Predictive Maintenance (PdM) using Tiny Machine Learning (TinyML) deployed on dual-core microcontrollers. As industrial assets in the Industry 4.0 era become increasingly complex, maintenance costs have grown to represent between 15% and 60% of total manufacturing expenses. This project shifts computational intelligence directly to the machine source, enabling autonomous, real-time diagnostics with sub-second response times. By integrating multi-modal sensing specifically vibration, electrical current, and thermal data the system identifies critical degradation patterns locally, ensuring operational reliability without the latency and privacy concerns of cloud-based architectures.

1.2 Background

The evolution of industrial maintenance can be traced through distinct philosophical generations, each aiming to improve the reliability and longevity of physical assets.

1.2.1 Types of Maintenance

- **Corrective Maintenance (Generation I):** Also known as reactive or "Run-to-Failure" maintenance, this approach involves repairing or replacing equipment only after an obvious failure has occurred. While it requires low initial investment, it frequently leads to unplanned downtime and secondary damage, costing approximately three times more than repairs performed before total failure.

- **Preventive Maintenance (Generation II):** This strategy is time-based or machine-run-based, involving scheduled inspections and part replacements at predetermined intervals regardless of the actual condition of the asset. Although it increases reliability over reactive models, it is often wasteful as functional components are frequently discarded too early.
- **Condition-Based Maintenance (CBM):** Often considered the third generation of maintenance, CBM is a non-invasive strategy that uses sensors to periodically or continuously monitor mechanical and operational parameters. The fundamental motto of CBM is "Do not fix what is not broken". It assumes that physical features such as vibration amplitudes, thermal buildup, and electrical load serve as measurable gauges of machine health.

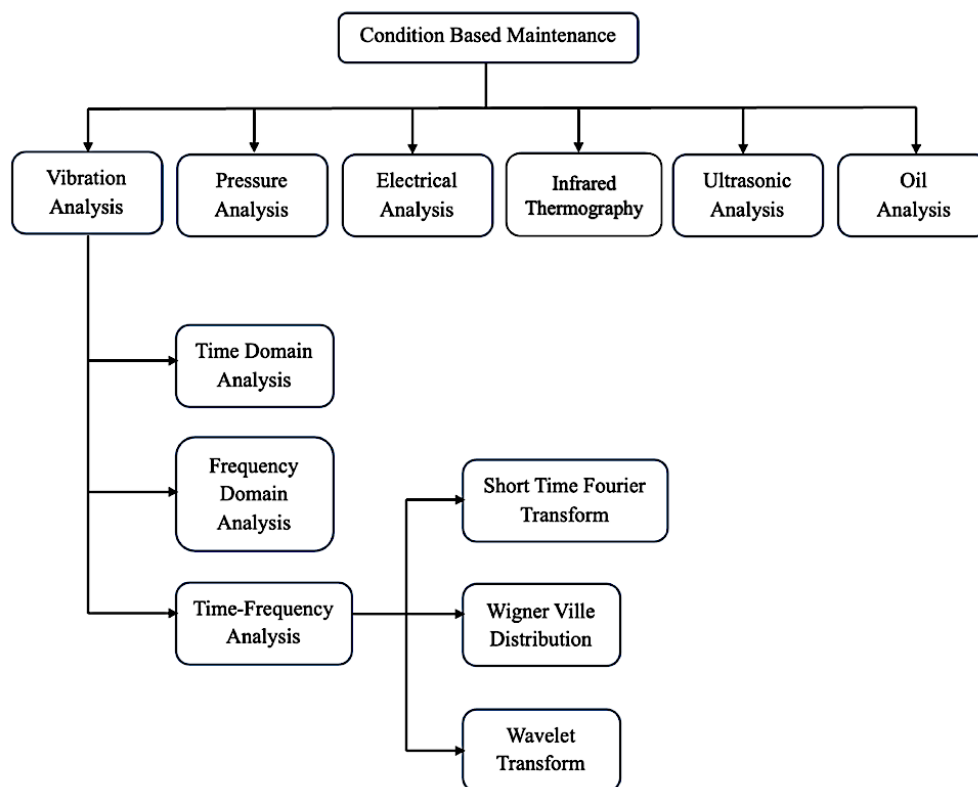


Fig. 1.1: Condition based maintenance

The Fig. 1.1 illustrates various Condition based maintenance techniques used for monitoring industrial machinery health. Among these methods, vibration analysis plays a major role in the proposed project for detecting faults in rotating equipment such as bearings and shafts. The diagram also highlights different signal analysis approaches including time-domain, frequency-domain, and time-frequency analysis techniques such as

Short-Time Fourier Transform (STFT), Wavelet Transform, and Wigner-Ville Distribution.

In the proposed Edge-Intelligent Condition Monitoring System, vibration signals acquired from sensors are processed using FFT-based frequency-domain analysis and TinyML algorithms for real-time fault diagnosis, predictive maintenance, and Remaining Useful Life estimation in Industrial IoT environments.

1.2.2 Edge-Intelligent Monitoring System

The proposed system represents a transition from traditional CBM to an Edge-Intelligent Monitoring System, which incorporates local computational robustness to process data at the extreme edge of the network.

- **Multi-Modal Data Fusion:** Unlike single-modality systems, an edge-intelligent framework integrates and correlates high-frequency triaxial vibration signals with non-invasive electrical current and thermal signatures. This correlation is critical for minimizing false positives; for instance, a thermal spike is only flagged as a critical fault if accompanied by abnormal spectral vibration or current consumption deviations.
- **Decentralized Inference:** By deploying computational power directly at the machine source, the system overcomes the high latency (typically $> 500\text{ms}$) and bandwidth constraints of cloud-based Predictive Maintenance. Intelligence is executed locally via a dual-core synchronization strategy where one core manages high-frequency signal acquisition while the second core handles real-time AI inference.
- **Prognostic Depth:** The system moves beyond "snapshot" anomaly detection which only identifies a fault after it has manifested to provide prognostic insights into the Remaining Useful Life of an asset. This is achieved through a noise-resilient pipeline utilizing Fast Fourier Transform for spectral feature extraction and 8-bit quantized deep learning models for local time-series history analysis.

1.3 Objectives of the Proposed Work

The primary goal of this research is to bridge the gap between simple anomaly detection and comprehensive prognostic monitoring at the edge through the following objectives:

1. To design and deploy a multi-modal TinyML framework on a dual-core microcontroller that fuses triaxial vibration, electrical current, and temperature data to classify mechanical faults (e.g., bearing wear, imbalance) and estimate the asset Remaining Useful Life.
2. To implement a noise-resilient preprocessing pipeline using the Fast Fourier Transform for spectral feature extraction, ensuring robust diagnostic performance in harsh, electromagnetically noisy industrial environments.

1.4 Hardware and Software Requirements

The implementation of the prognostic node requires industrial-grade sensing hardware and a specialized software stack to facilitate real-time edge intelligence.

1.4.1 Hardware Requirements

- **Microcontroller:** A dual-core 32-bit microcontroller (240 MHz, 520 KB SRAM) for simultaneous high-frequency data acquisition and AI inference.
- **Vibration Sensor:** A triaxial Micro-Electro-Mechanical Systems accelerometer to capture high-frequency vibrations indicating mechanical degradation.
- **Current Sensor:** A non-invasive current transformer (30A/50mA) to monitor electrical load and detect stator-related faults.
- **Temperature Sensor:** A waterproof, digital stainless-steel probe to monitor motor heat buildup and prevent thermal failure.
- **Control hardware:** An industrial **Variable Frequency Drive** regulates the speed of a **0.37 kW (0.5 Hp) induction motor**, enabling diagnostic validation across various operational RPMs.

1.4.2 Software Requirements

- **TinyML Pipeline:** A specialized edge AI platform for data labeling, training memory-optimized models, and implementing **8-bit quantization** to reduce the model's memory footprint.
- **Firmware Development:** An integrated development environment (C/C++ based) for developing embedded code to manage sensor interfaces and execute TinyML libraries.
- **Communication & Visualization:** The **MQTT protocol** is utilized for lightweight transmission of diagnostic health data to cloud-based IoT dashboards for real-time tracking.

Chapter 2

LITERATURE SURVEY

The literature survey was carried out to study recent developments in Edge-Intelligent Condition Monitoring Systems, Tiny Machine Learning (TinyML), vibration-based fault diagnosis, Industrial Internet of Things (IIoT), and predictive maintenance techniques for rotating machinery. The reviewed papers helped identify the limitations of existing cloud-based and single-modality monitoring systems and provided valuable insights for developing the proposed multi-modal edge-intelligent predictive maintenance framework.

Krishna et al. (2025) [1] developed an intelligent motor control and monitoring architecture leveraging the ESP32 microcontroller and Internet of Things technologies to facilitate the remote operation of multiple motor assets. The study demonstrates that utilizing the dual-core processing capabilities of the microcontroller allows for command latencies below 500ms and supports fundamental predictive maintenance through the real-time feedback of thermal and electrical consumption metrics.

Gupta and Shivhare (2025) [2] introduced a TinyML-based framework for real-time vibration analysis deployed directly on edge hardware. By implementing a memory-optimized 1D Convolutional Neural Network, the study achieved a diagnostic accuracy exceeding 92% in identifying specific mechanical faults, such as shaft misalignment and bearing degradation. The research concludes that shifting computational tasks from centralized cloud locations to the source of data acquisition is vital for achieving the sub-second response times required for industrial safety.

Zonta et al. (2020) [3] conducted a comprehensive systematic literature review was conducted to establish a taxonomy for Predictive Maintenance (within the Industry 4.0 ecosystem). The authors identified that maintenance operations represent a significant portion of manufacturing expenses, typically ranging from 15% to 60% of total operational costs. The findings emphasize a strategic industry-wide transition from reactive "fail-and-fix" strategies toward intelligent, data-driven Artificial Intelligence models.

Thakkar and Kumar (2024) [4] explored the integration of machine learning algorithms with edge computing to enable real-time anomaly detection and Remaining Useful Life estimation for industrial assets. The authors concluded that processing data near the source

effectively addresses the critical industrial constraints of network latency, excessive bandwidth consumption, and data privacy concerns.

Herrera et al. (2025) [5] detailed the implementation of a low-cost, open-source prototype for bearing failure detection using a frequency-domain analysis pipeline. The system utilized an end-to-end workflow consisting of vibration signal capture, Fast Fourier Transform spectral analysis, and a Multilayer Perceptron classifier executed on an embedded platform. This research confirms that affordable, resource-constrained hardware can successfully host sophisticated intelligence for industrial state identification.

Ooko and Karume (2024) [6] provided a systematic review was provided on the application of TinyML for industrial maintenance, highlighting its role in enhancing system responsiveness. The research demonstrates that on-device inference significantly reduces the energy consumption typically required for the high-bandwidth wireless transmission of raw sensor data to cloud servers.

Liu et al. (2018) [7] reviewed Artificial Intelligence techniques for fault diagnosis in rotating machinery. The research highlighted the effectiveness of deep learning, signal processing, and vibration analysis methods for identifying bearing faults, shaft imbalance, and other rotating machinery defects under complex operating conditions.

Katib et al. (2025) [8] proposed a TinyML-driven real-time anomaly detection framework for IoT consumer devices. The study demonstrated how deep learning models deployed on edge devices improve predictive maintenance performance while enhancing data security and reducing communication overhead in distributed IoT systems.

Kumar et al. (2018) [9] investigated fault diagnosis in belt drive transmission systems using wavelet spectrum analysis. The study proved that time-frequency signal analysis techniques effectively identify mechanical faults and dynamic vibration abnormalities in rotating transmission systems.

Sridhara et al. (2018 –19) [10] developed a MATLAB-based signal processing approach for bearing fault detection. The work demonstrated the importance of vibration signal analysis and statistical indicators such as Root Mean Square and kurtosis for identifying bearing defects.

Raghavendra [11] developed an experimental bearing test rig for collecting vibration data for automatic fault identification. The study established the importance of controlled experimental setups for generating fault datasets required for training machine learning-based predictive maintenance systems.

2.1 Key Insights

An analysis of the collected literature reveals several critical technical insights regarding the development of edge-native condition monitoring systems. These insights establish the architectural requirements for achieving high diagnostic fidelity within resource-constrained industrial environments.

- **Edge-Native Computational Superiority:** Shifting computational intelligence from centralized cloud architectures directly to the machine source enables sub-second inference latency (targeted below 250ms), effectively overcoming the bandwidth consumption, unpredictable network delays, and data privacy risks associated with traditional cloud-streaming systems.
- **Modality and Noise Resilience:** While vibration analysis is the primary diagnostic modality for rotating machinery, frequency-domain preprocessing via the Fast Fourier Transform is essential for isolating fault-specific harmonic components from the heavy electromagnetic and mechanical noise prevalent in industrial settings.
- **Resource-Aware Model Optimization:** To deploy sophisticated deep learning architectures on microcontrollers with limited Static Random-Access Memory and flash storage, the implementation of 8-bit integer quantization and model pruning is mandatory to reduce the memory footprint without incurring substantial losses in diagnostic accuracy.

- **Multi-Modal Fusion for Diagnostic Fidelity:** The integration and correlation of multi-modal data specifically triaxial vibration, non-invasive electrical current, and thermal readings significantly improve the robustness of the diagnostic engine, minimizing false-positive alerts and providing the time-series history necessary for accurate Remaining Useful Life estimation.

3.3 Research Gap Identification

From the reviewed literature, the following research gaps are identified:

- **Prognostic Depth Remaining Useful Life:** Existing research implementations frequently treat predictive maintenance as a "snapshot" alert system detecting a fault only after it has manifested rather than providing the comprehensive time-series histories required to estimate the Remaining Useful Life locally on the edge device.
- **Integrated Multi-Modal Fusion:** There is a critical deficiency in end-to-end, single-chip systems that simultaneously integrate and correlate multi-modal data specifically triaxial vibration, non-invasive electrical current, and temperature to differentiate between similar mechanical and electrical fault signatures and minimize false positives.

CHAPTER 3

PROBLEM DEFINITION

Development of an Edge-Intelligent Condition Monitoring System for rotating machinery to enable real-time fault detection, predictive maintenance, and Remaining Useful Life estimation while minimizing unplanned downtime, maintenance cost, and dependency on cloud-based monitoring systems.

3.1 Problem Statement

Industrial rotating machinery constitutes the core of modern production lines, yet these assets remain inherently vulnerable to mechanical and electrical failures that lead to catastrophic downtime, significant financial losses, and personnel safety risks. Current maintenance strategies are frequently suboptimal: reactive maintenance results in costly, unplanned shutdowns, while preventive maintenance often leads to the wasteful replacement of functional components based on rigid schedules.

While Predictive Maintenance (PdM) offers a superior alternative, existing frameworks primarily rely on cloud-based architectures, which introduce critical technical barriers:

Latency: Network delays frequently exceed 500ms, making cloud systems unsuitable for real-time safety cut-offs during rapid component failure.

Bandwidth: Streaming high-frequency triaxial vibration data from multiple machine nodes consumes excessive network resources.

Security: Transmitting sensitive operational data to external servers raises significant industrial privacy concerns.

Furthermore, industrial environments are characterized by heavy electromagnetic interference and mechanical background noise, which contaminate sensor data and lead to false-positive fault detection. Most low-cost edge implementations lack the computational robustness to perform sophisticated noise filtering, spectral analysis, and multi-modal data fusion locally. Consequently, there is a critical absence of an integrated, edge-native solution that can accurately classify faults and provide prognostic insights, such as Remaining Useful Life estimation, within the strict resource constraints of a microcontroller.

3.2 Scope of the Project

The scope of this research involves the development and validation of an autonomous prognostic node designed for industrial induction motors. The system is engineered to perform the following technical operations locally on a single-chip microcontroller:

Multi-Modal Acquisition: Simultaneous sampling of high-frequency triaxial vibration, non-invasive electrical current, and thermal signatures.

Edge Preprocessing: Execution of an embedded Fast Fourier Transform pipeline and Z-score normalization to isolate fault-related spectral signatures from background noise.

State Classification: Identification of specific machine states, including mechanical imbalance, shaft misalignment, and rolling-element bearing wear, using optimized convolutional architectures.

Actionable Intelligence: Implementation of a dual-output support layer that triggers local hardware safety responses via a Variable Frequency Drive and updates a remote IoT dashboard for long-term prognostic oversight.

3.3 Constraints and Assumptions

This subsection delineates the technical boundaries and the operational premises upon which the edge-intelligent system is architected. It defines the hardware-specific limitations and the fundamental industrial conditions assumed for the reliable execution of the TinyML diagnostic framework.

3.3.1 Constraints:

- **Memory and Computational Capacity:** To ensure compatibility with the ESP32-WROOM-32 microcontroller, all deep learning architectures must be optimized via 8-bit integer quantization and model pruning to operate within the 520 KB of Static Random-Access Memory and limited flash storage.
- **Real-Time Latency:** To facilitate an effective autonomous safety cutoff through the control hardware, the end-to-end inference latency spanning from multi-modal data

acquisition and FFT-based feature extraction to final state classification is strictly limited to a sub-second interval, targeted below 250 milliseconds.

- **Hardware Economic Viability:** The total prototype implementation cost is constrained to approximately (utilizing an industrial-grade motor and starter configuration) to remain a cost-effective alternative for small-to-medium industrial enterprises.

3.3.2 Assumptions:

- **Correlated Physical Indicators:** It is technically assumed that nearly all mechanical and electrical failures in industrial induction motors are preceded by correlated physical indications specifically triaxial vibration shifts, localized thermal buildup, or current consumption deviations detectable by the sensor suite during incipient stages.
- **Predictable Failure Progressions:** The system operates on the assumption that machinery exhibits identifiable degradation signatures (such as harmonic spikes at specific fault frequencies) for a sufficient temporal duration to allow the TinyML model to provide prognostic alerts before a functional failure occurs.
- **Operational Regime Consistency:** For the validation of diagnostic accuracy, it is assumed that the induction motor operates within the Variable Frequency Drive regulated speed regimes where the mechanical load is sufficiently stationary to avoid sensor saturation or excessive signal clipping during spectral analysis.

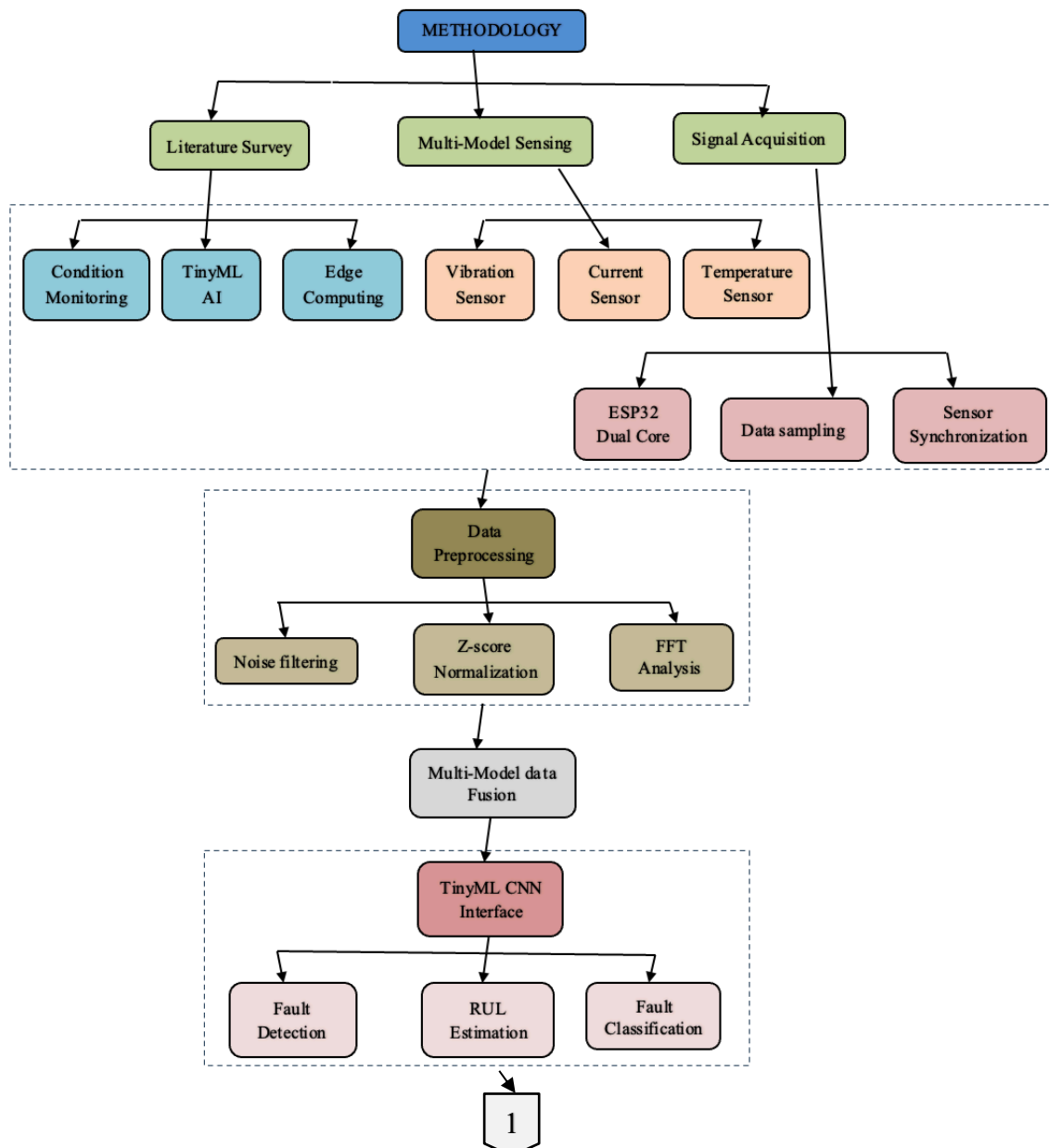
CHAPTER 4

PROPOSED METHODOLOGY

This chapter describes the technical architecture and operational framework of the proposed Edge-Intelligent Condition Monitoring System. It explains the integration of multi-modal sensors, noise-resilient signal preprocessing techniques, and TinyML-based neural network deployment on dual-core microcontrollers for autonomous real-time fault diagnosis and predictive maintenance.

4.1 Proposed Approach and Algorithms

The framework adopts a multi-modal data fusion strategy to overcome the diagnostic limitations of single-modality vibration monitoring. By correlating mechanical (vibration),



electrical (current), and thermal (temperature) indicators, the system minimizes false-positive alerts and improves the robustness of the diagnostic engine.

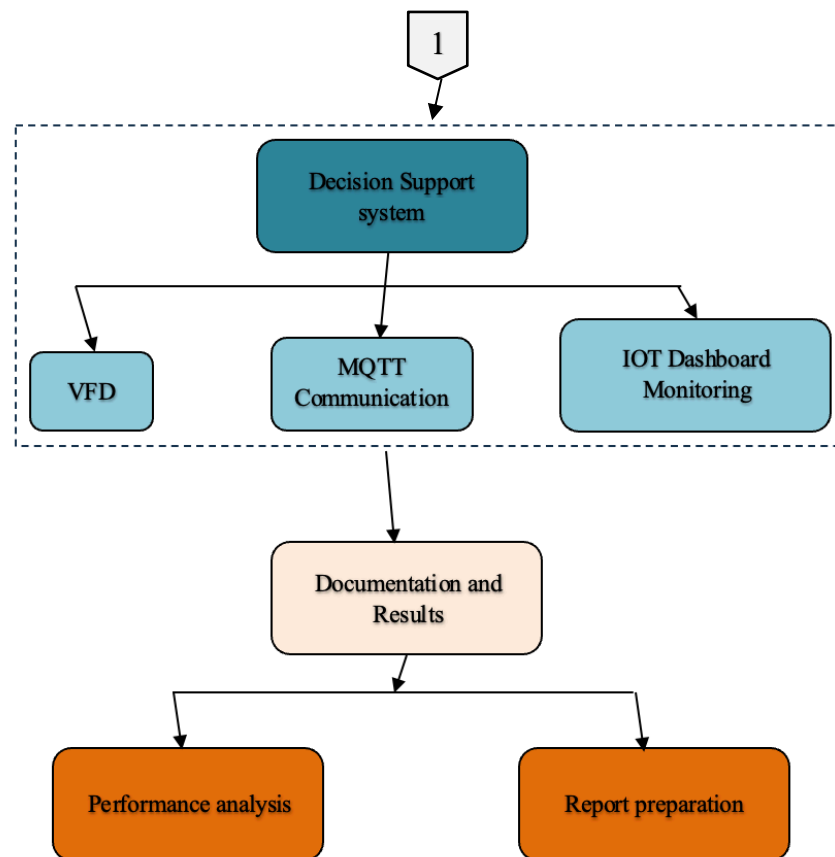


Fig. 4.1: Methodology flow chart

Fig. 4.1 Methodology flowchart of the proposed Edge-Intelligent Condition Monitoring System illustrating literature survey, multi-modal sensor data acquisition, signal preprocessing using FFT and Z-score normalization, TinyML-based fault classification, Remaining Useful Life estimation, and IoT-enabled decision support for real-time predictive maintenance.

Noise-Resilient Spectral Preprocessing: To ensure operational fidelity in electromagnetically noisy industrial environments, raw time-domain signals are subjected to an embedded Fast Fourier Transform pipeline. This algorithm converts stochastic time-series data into identifiable spectral signatures, isolating mechanical fault harmonic components such as bearing impact frequencies from background environmental interference. Preceding frequency-domain conversion, the signals undergo Z-score normalization to ensure uniform data magnitudes for the neural network input layer.

Quantized Convolutional Neural Architecture: The diagnostic core utilizes a 1D Convolutional Neural Network specifically optimized for spatial feature extraction from multi-axis vibration signals. To comply with the strict resource constraints of the ESP32 microcontroller, the model is subjected to 8-bit integer quantization and weight pruning. This optimization reduces the memory footprint to fit within the 520 KB Static Random-Access Memory while maintaining sub-second inference latency.

Multi-Modal Correlation and Prognostics: The system moves beyond snapshot anomaly detection by implementing time-series regression for Remaining Useful Life estimation. The algorithm correlates thermal buildup with high-frequency spectral deviations; for instance, a temperature spike is only classified as a critical bearing fault if accompanied by a corresponding increase in specific spectral energy bands.

4.2 Techniques and Tools to be Used

The implementation utilizes an integrated industrial-grade hardware-software stack designed for real-time edge intelligence.

4.2.1 Hardware Instrumentation:

- **Central Processing Unit:** The ESP32-WROOM-32 DevKit is utilized, leveraging its dual-core architecture to synchronize high-frequency data acquisition and AI inference simultaneously.
- **Sensing Suite:** Orthogonal mechanical vibrations are captured via a triaxial MEMS accelerometer (MPU6050). Non-invasive electrical load monitoring is performed by an SCT-013-000 current transformer. Thermal signals are acquired using a digital DS18B20 waterproof probe.
- **Control and Asset Unit:** A 0.37 kW Variable Frequency Drive (VFD) regulates a three-phase induction motor, enabling the system to validate diagnostic accuracy across various operational RPMs.

4.2.2 Software Development Frameworks:

- **TinyML Pipeline:** The Edge Impulse platform is utilized for multi-modal data labeling, training memory-optimized models, and generating 8-bit quantized C/C++ libraries.
- **Firmware and Communication:** Firmware is developed using the Arduino IDE environment. Diagnostic results and health status packets are transmitted via the MQTT protocol to cloud-based IoT dashboards for prognostic oversight.

4.3 System Design and Workflow

The system is architected as a modular six-layer Industrial IoT framework designed for autonomous operation:

Layer 1: Sensing Layer: Continuous acquisition of raw multi-modal signals from the rotating asset.

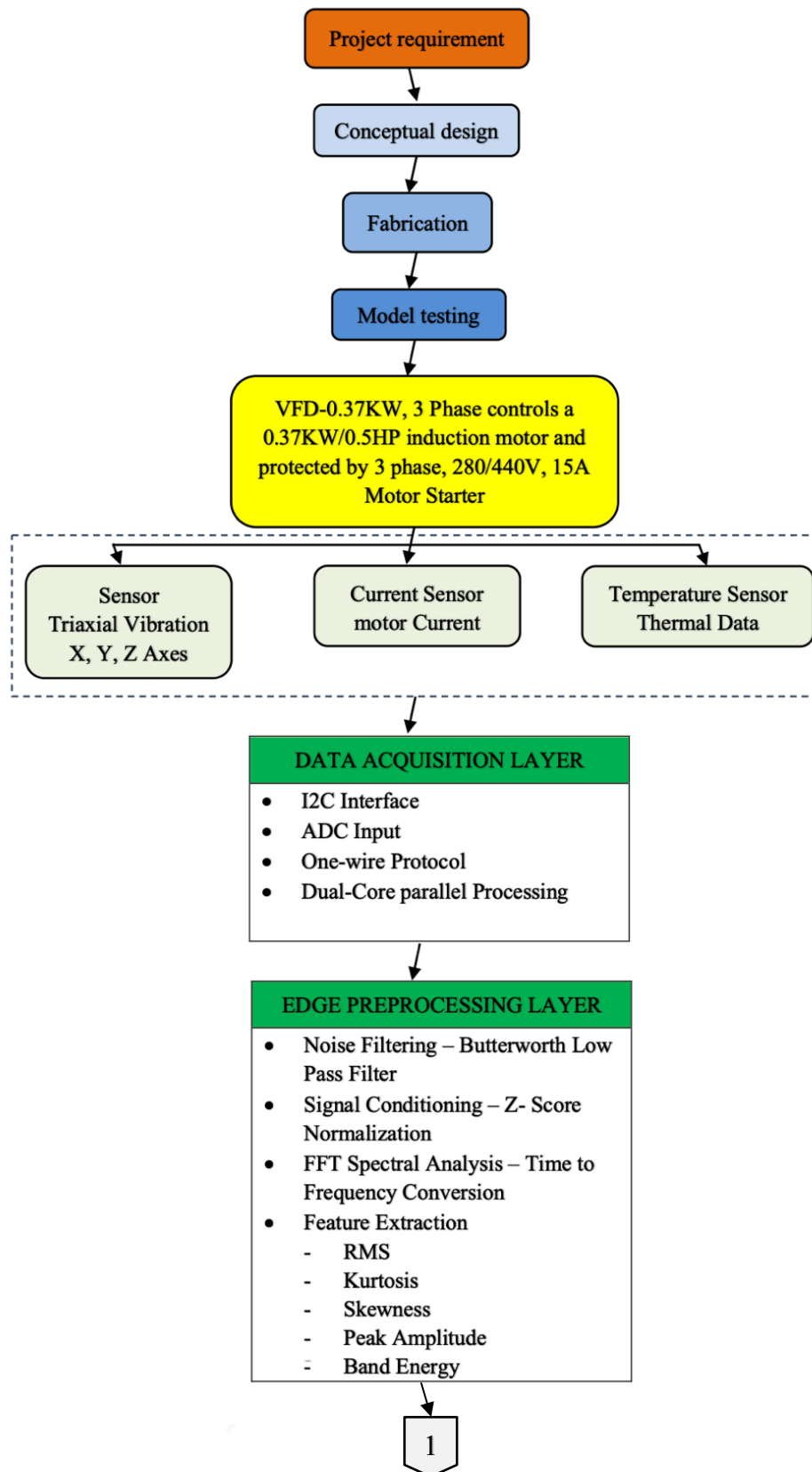
Layer 2: Data Ingestion Layer: Secure handling of sampled data using lightweight communication protocols for edge synchronization.

Layer 3: Processing Layer (Edge Preprocessing): Execution of noise filtering, Z-score normalization, and FFT conversion locally on the ESP32.

Layer 4: Model Repository Layer: On-device storage of optimized 8-bit quantized 1D CNN models.

Layer 5: Inference Layer (Edge Intelligence): Real-time execution of the diagnostic engine to classify machine states and estimate RUL.

Layer 6: Decision Support Layer: Activation of local hardware responses (safety cutoff via VFD) and remote visualization on IoT dashboards.



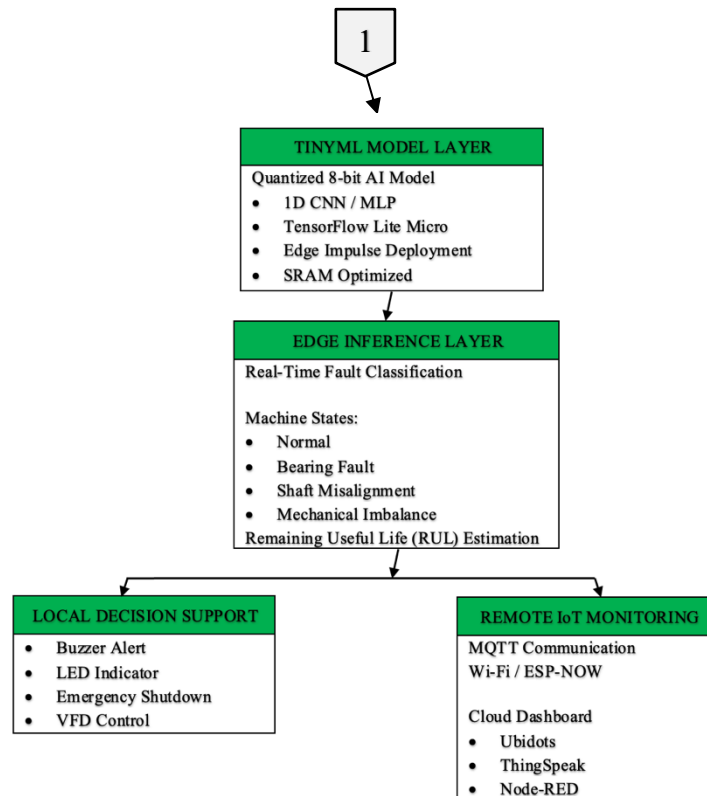


Fig 4.2: Six-layer Industrial IoT

Fig. 4.2 Six-layer Industrial IoT (IIoT) framework for autonomous edge-intelligent condition monitoring, illustrating sensing, data acquisition, edge preprocessing, model storage, TinyML-based inference, and decision support layers for real-time predictive maintenance and IoT-based monitoring.

CHAPTER 5

SUMMARY AND FUTURE PLAN

This chapter provides a comprehensive technical synthesis of the engineering milestones achieved during the preliminary phase of the research. It delineates the systematic transition from theoretical literature analysis and problem identification to the successful fabrication and validation of an edge-intelligent experimental test rig. Furthermore, this section establishes a strategic roadmap for Phase 2, focusing on advanced prognostic modeling and industrial-grade system hardening.

5.1 Summary of Work Completed in Phase 1

The primary focus of Phase 1 was to establish the architectural foundation and experimental validity of the Edge-Intelligent Condition Monitoring System. This phase successfully bridged the gap between institutional foundational research and modern edge-native AI deployment.

1. **Literature Survey and Gap Analysis:** Conducted a comprehensive technical analysis of existing predictive maintenance (PdM) frameworks, identifying a critical deficiency in end-to-end, single-chip edge systems that suffer from excessive bandwidth consumption and cloud latencies ($>500\text{ms}$). (26 February – 2 March)
2. **Problem Definition Formulation:** Finalized the core problem statement focusing on the absence of a noise-resilient, resource-constrained solution capable of executing local spectral analysis and multi-modal feature extraction. (3 March – 18 March)
3. **Industrial IIoT System Architecture Planning:** Formulated a modular six-layer Industrial IoT (IIoT) architecture to structure data flow seamlessly from raw physical sensing up to localized edge decision support. (10 March – 20 March)
4. **Hardware Stack Selection:** Evaluated and selected an optimized TinyML hardware stack featuring the dual-core ESP32-WROOM-32 microcontroller, MPU6050 triaxial accelerometer, SCT-013-000 current sensor, and DS18B20 thermal probe. (18 March – 25 March)

5. **Experimental Test Rig Design & Fabrication:** Fabricated a professional-grade experimental model test rig utilizing a 0.37 kW (0.5 Hp) three-phase induction motor, 15A motor starter, and a Schneider Variable Frequency Drive (VFD) for precise rotational speed control. (26 March – 5 April)
6. **Fault Emulation & Dataset Generation:** Successfully emulated controlled mechanical faults (shaft misalignment, structural unbalance) across different operational regimes to collect high-frequency spectral datasets. (6 April – 12 April)
7. **TinyML Framework Optimization:** Formulated and benchmarked the initial 8-bit post-training quantized 1D Convolutional Neural Network (CNN) architecture optimized for embedded execution within the memory constraints of the ESP32. (13 April – 20 April)

5.2 Plan for Phase 2 Implementation

Phase 2 focuses on evolving the current diagnostic prototype into a robust industrial prognostic node with advanced forecasting capabilities. The implementation is scheduled over a four-month duration (Months 5 – 8) to ensure technical depth and validation.

1. **Prognostic Regression Modeling (RUL Estimation):** Develop and train time-series regression models, specifically quantized hybrid CNN-LSTM architectures, to shift the system from fault classification to local Remaining Useful Life (RUL) estimation. (May - June)
2. **Acoustic Sensor Array Integration:** Integrate an advanced acoustic sensing suite utilizing MEMS microphones to capture early-stage frictional and aerodynamic anomalies before they translate into structural vibrations. (June - July)
3. **Energy Optimization & Intermittent Computing:** Program and implement advanced ESP32 power management profiles (light and deep sleep routines) alongside intermittent computing strategies to support decentralized, long-term autonomous operations. (July)

4. **Industrial Protocols Hardening:** Upgrade and secure the data ingestion layer to comply with standardized industrial protocols, including secure MQTT and OPC UA, facilitating edge-to-factory networking. (July - August)
5. **On-Device Continuous TinyML Evaluation:** Deploy the optimized multi-modal fusion model to the microcontroller and evaluate real-time sub-second inference latency, memory footprint execution, and algorithmic edge accuracy. (August)
6. **Real-World Factory Ecosystem Validation:** Install the integrated prognostic node onto industrial machinery to validate overall diagnostic fidelity, noise resilience, and operational efficiency under harsh factory conditions. (September)

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APPENDICES

A. Experimental Model Testing Rig



Fig A.1: Experimental Model Testing Rig

The Fig A.1 shows the experimental prototype developed for the proposed **Edge-Intelligent Condition Monitoring and Predictive Maintenance System**. The setup consists of an induction motor coupled with a pulley-belt transmission mechanism and rotating shaft supported by rolling element bearings. The rig is designed to simulate industrial rotating machinery conditions for real-time fault diagnosis and predictive maintenance applications. The induction motor drives the shaft through the V-belt arrangement, while the bearing assembly acts as the primary component for monitoring mechanical health conditions such as vibration abnormalities, shaft misalignment, imbalance, and bearing wear. Multi-modal sensors including vibration, current, and temperature sensors are integrated with the setup to continuously acquire machine condition data. The acquired signals are processed using FFT-based spectral analysis and TinyML-based edge inference on the ESP32 platform for autonomous fault classification and Remaining Useful Life (RUL) estimation. The experimental rig enables testing under different loading conditions to evaluate the effectiveness of the proposed Industrial IoT (IIoT)-enabled predictive maintenance framework.

MOTOR

- Capacity: 3-Phase, 440V, 50Hz
- Power: 0.37KW, 0.5HP
- Rated speed: 1100 RPM
- Rated current: 1.4 Amps
- Amb. Temperature: 50° C

MOTOR STARTER

- Phase: 3
- Coil Voltage: 280/440V
- Amperes: 15 Amps
- Cycles: 50
- Relay Range: 13-21

B. 2D Conceptual Design

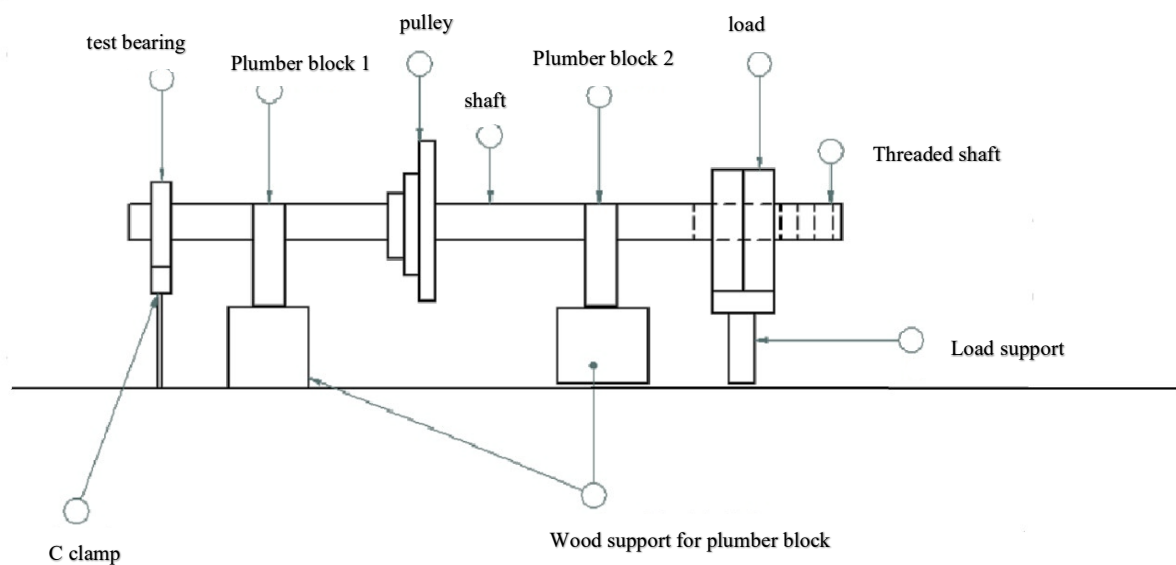


Fig B.2: 2D conceptual design

The Fig B.2 illustrates the 2D conceptual design of the proposed **Edge-Intelligent Condition Monitoring and Predictive Maintenance System** developed for rotating machinery analysis. The setup consists of a rotating shaft supported by Plummer block bearings, a pulley-driven transmission system, and a controllable loading mechanism mounted on a rigid base structure.

In the experimental arrangement, the shaft is coupled to the driving motor through a pulley-belt mechanism to simulate real industrial machine operation. The test bearing mounted at one end of the shaft acts as the primary monitoring component for analyzing vibration signatures under healthy and faulty operating conditions. Plummer blocks provide shaft support and maintain rotational stability during experimentation.

A threaded loading mechanism with external load support is incorporated to apply variable mechanical loading conditions such as no-load, half-load, and full-load operation. This enables the proposed system to study the effect of dynamic loading on bearing vibration characteristics and fault propagation.

The conceptual setup is integrated with multi-modal sensors including vibration, current, and temperature sensors interfaced with the ESP32 edge controller. The acquired signals are processed using FFT-based feature extraction and TinyML algorithms for real-time fault classification, anomaly detection, and Remaining Useful Life estimation within the Industrial IoT framework.